Continual Vision-and-Language Navigation Supplementary Material

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1 Dataset Analysis

We examine the existence of distribution shifts across scene domains by calculating the Fréchet Inception Distance (FID) between visual and textual features across scene domains [6]. In Initial-instruction based CVLN (I-CVLN), 10 out of 20 scene domains are selected randomly, and in Dialogue based CVLN (D-CVLN), 10 out of 11. We use Resnet-152 trained on Place365 for visual feature extraction in I-CVLN [7, 9], and ViT-B/16 for D-CVLN [4]. These analyses are visualized as heatmaps in Figure 1, confirming significant visual feature distribution shifts in both datasets. For textual features in both I-CVLN and D-CVLN, we utilize BERT for extraction [3], with differences across scene domains examined through FID, visualized in Figure 2.

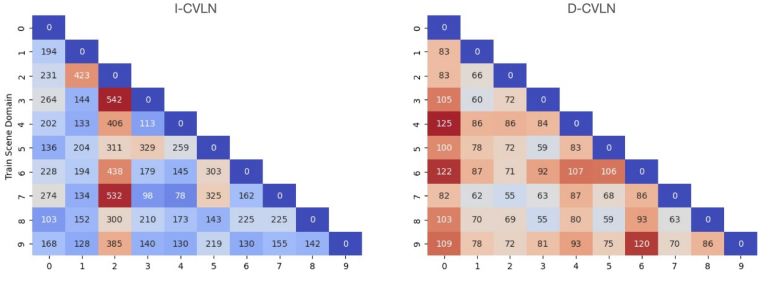


Figure 1: Heatmaps illustrating the distribution shifts between visual features across scene domains for the I-CVLN and D-CVLN.

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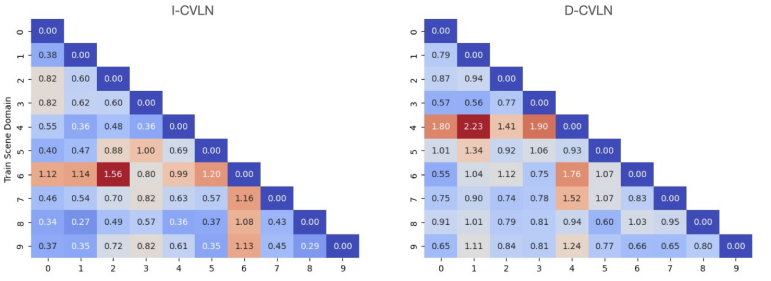


Figure 2: Heatmaps representing the distribution shifts between textual features across scene domains within the I-CVLN and D-CVLN datasets.

To further investigate textual distribution differences, we generate word clouds from texts associated with five scene domains for I-CVLN and D-CVLN, visualized in Figure 3. These word clouds highlight the lexical commonalities and limited diversity within navigation-related texts.

2 Training Details

For training I-CVLN and D-CVLN, we adhere to training methodologies from previous works, mixing Reinforcement Learning (RL) and Imitation Learning (IL) objectives [1, 7]. A3C is employed for RL [8], and for IL, the agent calculates cross-entropy loss for each action by following teacher actions to traverse the ground-truth trajectory.

3 Implementation Details

We utilize transformer-based models, VLN-BERT for I-CVLN [7] and HAMT for D-CVLN [1], with training on NVIDIA A100 GPUs. Batch size is set at 8, with 4,000 iterations per scene domain for I-CVLN and 5,000 for D-CVLN. Image encoding employs ResNet-152 and ViT-B/16 for I-CVLN and D-CVLN, respectively [4, 5, 9].



Figure 3: Word clouds for five selected scene domains from the D-CVLN datasets.Word clouds depicting the textual distribution within five selected scene domains of the I-CVLN dataset.

4 Additional Method Details

4.1 Perplexity Replay Algorithm Pseudocode

Algorithm 1 Perplexity Replay Memory Update

Input: Replay memory M, replay memory size $|M|$, set of episodes from the last scene domain E_{last}, agent parameters θ, the number of trained scene domains s
Output: Updated replay memory M
Calculate Action Perplexity (AP) for each episode in E_{last} and M
$E_{\text{last}} \leftarrow \text{sort}(E_{\text{last}}, \text{by=AP, descending})$
$M \leftarrow \text{sort}(M, \text{by=AP, ascending})$
$n_s \leftarrow |M|/(s+1)$
for $s = 1$ to s **do**
    Remove episodes with the lowest AP from M for domain s until only n_s episodes remain
end for
Add episodes from E_{last} to M until $|M|$ equals fixed size
return M

4.2 Episodic Self-Replay Algorithm Pseudocode

Algorithm 2 Episodic Self-Replay

Require: Dataset D_s, Memory M, Learning rate η, Agent f_θ, GetObservation(t), ActionExecuter(a_t), λ_1, λ_2

Sample $(I, A^*) \sim D_s$
$\mathcal{L}_{D_c} \leftarrow 0$, $Z \leftarrow []$, $t \leftarrow 1$
$v_t \leftarrow$ GetObservation(t);
while a_t does not predict `stop` **do**
  $z_t \leftarrow f_\theta(I, v_t)$; Z.append(z_t)
  $a_t \leftarrow \arg\max(z_t)$
  $a_t^* \leftarrow A^*(t)$ //Get true action
  $\mathcal{L}_{D_c} \leftarrow \mathcal{L}_{D_c} + \ell(a_t, a_t^*)$
  $v_{t+1} \leftarrow$ ActionExecuter(a_t)
  $t \leftarrow t+1$
end while
$\mathcal{L}_{D_c} \leftarrow \mathcal{L}_{D_c}/t$
Update $M \leftarrow$ Random$(M, (I, A^*, Z))$
Sample $(I', A'^*, Z'^*) \sim M$
$\mathcal{L}_{ESR} \leftarrow 0$, $\mathcal{L}_M \leftarrow 0$, $t \leftarrow 1$
$v_t \leftarrow$ GetObservation(t);
while a_t does not predict `stop` **do**
  $z_t \leftarrow f_\theta(I', v_t)$
  $a_t \leftarrow \arg\max(z_t)$
  $a_t'^* \leftarrow A'^*(t)$ //Get true action
  $\mathcal{L}_M \leftarrow \mathcal{L}_M + \ell(a_t, a_t'^*)$
  $z_t'^* \leftarrow Z'^*(t)$ //Get replay logit
  $\mathcal{L}_{ESR} \leftarrow \mathcal{L}_{ESR} + \|z_t - z_t'^*\|_F^2$ //Logit distillation
  $a_t^* \leftarrow \arg\max(z_t'^*)$ //Get replay logit's action
  $v_{t+1} \leftarrow$ ActionExecuter(a_t^*)
  $t \leftarrow t+1$
end while
$\mathcal{L}_{ESR} \leftarrow \mathcal{L}_{ESR}/t$
$\mathcal{L}_M \leftarrow \mathcal{L}_M/t$
$\mathcal{L}_{ESR_total} \leftarrow \mathcal{L}_{D_c} + \lambda_1 \mathcal{L}_{ESR} + \lambda_2 \mathcal{L}_M$
$\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}_{ESR_total}$

5 Limitation and Future Work

The Continual Vision-and-Language Navigation (CVLN) paradigm represents a initial step forward in mitigating catastrophic forgetting through simple yet effective methods like Perplexity Replay and Episodic Self-Replay. However, it faces limitations in its current focus on indoor environments and the static approach to replay memory construction and exploitation. Future work should aim to expand the environmental scope to include complex outdoor settings, requiring the development of new datasets and models capable of navigating

the unpredictable and variable nature of outdoor spaces. Additionally, exploring dynamic and intelligent replay memory management strategies could enhance the balance between knowledge retention and adaptation to new information, making the models more efficient and scalable.

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